COMPILER DESIGN LAB

50% PROJECT REPORT

Compiler-Style Lexical Analysis and Character Popularity Modeling of Comic Dialogue Text

Team No: 11 |
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Abstract

Comic digitization pipelines process scanned physical pages through panel extraction, speech balloon segmentation, and Optical Character Recognition (OCR), resulting in unstructured dialogue strings. Traditional Natural Language Processing (NLP) and compiler-based text tools frequently struggle with comic dialogue due to unique typographical quirks: all-capitals lettering, ubiquitous onomatopoeia (e.g., 'BOOM', 'AAARGH'), ellipses ('...'), interrobangs ('?!'), and OCR noise leftovers (e.g., stray symbols such as '~' or '#'). In this work, we propose a robust compiler front-end pipeline that treats comic dialogue as a specialized micro-language while eliminating fragile Context-Free Grammar (CFG) syntax parsers that traditionally suffer from catastrophic rejection rates on non-standard spoken dialogue. The pipeline comprises Speaker Splitting, Normalization, an extended Comic Lexical Analyzer, Symbol Table management, Rule-Based Sentence Classification, and an integrated Character Popularity Engine. By leveraging lexical vocative patterns ('comma + name + terminator') and in-text mention discovery, the system resolves aliases to canonical entities and computes a multi-factor weighted prominence score. Evaluated on benchmark dialogue and 6,335 lines from 18 superhero movie scripts in the Cornell Movie-Dialogs Corpus, the proposed system achieves a 99.8% processing throughput rate at 0.098 ms/sentence—representing a 187x latency speedup over syntax parsing—while accurately extracting character prominence hierarchies and OCR error distributions.

Keywords: Comic Dialogue, Lexical Analysis, Compiler Front-End, Symbol Table, Vocative Addressing, Character Popularity Modeling, OCR Error Detection.

1. Introduction

Comic books and graphic novels represent a vital multimodal medium where narrative storytelling unfolds through sequential graphical panels and speech balloons. With the mass migration of readers to mobile devices, tablets, and digital archiving systems, automated comic digitization has become essential for content search, automatic narrative summarization, translation, accessibility text-to-speech for visually impaired individuals, and script generation. The base workflow for comic digitization typically involves image thresholding, panel contour extraction, speech balloon cropping, and text transcription via OCR engines such

as Tesseract. However, once OCR extracts raw text strings, traditional natural language systems face severe structural limitations.

Comic dialogue possesses significant stylistic idiosyncrasies that differentiate it from formal prose:
• Typography & Casing: Text is predominantly rendered in uppercase lettering, neutralizing case-based Named Entity Recognition (NER).
• Expressive Punctuation: Dramatic pacing frequently employs multiple ellipses ('...'), interrobangs ('?!'), and emphatic terminators ('!!!').
• Onomatopoeia & Elongation: Action and pain cues are expressed through sound effects ('BOOM', 'POW') and elongated screaming ('AAARGH', 'NOOOO').
• OCR Artifacts: Imperfect scanning, halftone ink bleed, and contour clipping leave stray non-alphanumeric noise ('~', '#', '|').

Original compiler-style approaches attempted to parse comic dialogue using strict Context-Free Grammars (CFGs). However, because dialogue naturally exhibits grammatical fragmentation, conversational ellipsis, and non-canonical syntax, CFG parsers reject between 60% and 75% of perfectly valid comic utterances. In this project, we deliberately eliminate the brittle syntactic parser. Instead, we design a high-throughput compiler front-end pipeline focused on robust lexical categorization, symbol table accumulation, heuristic sentence classification, and a first-class Character Popularity Module that determines character prominence through direct dialogue agency, vocative address, narrative mentions, and tonal cues.

2. Literature Survey

The automated processing of comic dialogue sits at the intersection of computer vision, compiler design, and computational narrative analysis. To contextualize our methodology, we review existing research across three major domains spanning 32 reference works.

2.1 Comic Digitization, Speech Balloon Detection, and OCR Text Extraction

Early comic digitization research focused on geometric image processing and visual heuristic segmentation. Ong and Ma [1] utilized oriented pattern analysis to segment comic panels from uniform gutters. Augereau et al. [2] and Matsui et al. [3] established comprehensive surveys on comic document analysis, identifying panel sequencing and balloon segmentation as foundational hurdles. With the advent of deep learning, Dubray and Laubrock [4] implemented deep convolutional neural networks (CNNs) for balloon segmentation, demonstrating that speech balloons exhibit varied elliptical and ragged geometries. Rigaud et al. [5], [6] integrated active contours and typography extraction to segment text lines within speech balloons. Lenadora et al. [7] developed an end-to-end framework extracting speech bubbles and running Tesseract OCR, but observed that raw transcriptions suffered from orphan punctuation and stray contour artifacts. Kato and Harada [8] and Qin et al. [9] demonstrated that OCR character error rates in graphic novels regularly exceed 15% to 22% due to custom display typography. These studies emphasize that post-OCR processing cannot assume clean, grammatical input and must explicitly isolate OCR noise.

2.2 Compiler Front-End Methodologies and Lexical Processing for Informal Text

Classical compiler theory, articulated by Aho, Lam, Sethi, and Ullman [10] and Cooper and Torczon [11], establishes that the front-end is responsible for linear character scanning, token classification, and symbol table management. While formal programming languages benefit from determinism, applying compiler techniques to natural language was explored by Jurafsky and Martin [12] and Grune et al. [13]. However, syntactic context-free parsing (e.g., Earley [14], Cocke-Younger-Kasami [15]) enforces strict acceptance criteria. In dialogue corpora, Manning et al. [16] and Ritter et al. [17] showed that informal social dialogue causes CFG acceptance rates to drop precipitously. Foster [18] and Geman and Johnson [19] documented that robust NLP requires probabilistic or shallow lexical methods rather than rigid syntax checking. Gimpel et al. [20] developed

tokenization strategies for informal micro-texts, noting that emoticons, punctuation clusters, and phonological lengthening (e.g., 'soooo') function as distinct lexical morphemes. In our work, we build on finite-state lexical analysis principles [21], [22] to capture comic tokens without imposing brittle grammar rejection rules.

2.3 Character Extraction, Social Network Analysis, and Prominence Modeling in Scripts

Extracting narrative networks and character prominence from creative texts has evolved significantly over the past decade. Elson, Dames, and McKeown [23] pioneered automated social network extraction from 19th-century novels based on conversational dialogue. Danescu-Niculescu-Mizil and Lee [24] introduced the Cornell Movie-Dialogs Corpus, providing over 304,000 conversational exchanges and illustrating that interpersonal linguistic alignment reflects character power dynamics. Agarwal et al. [25] analyzed character networks in literature by identifying direct speech and vocative interactions. Sang et al. [26] and Valls-Vargas et al. [27] demonstrated that character coreference resolution is critical because protagonists are referred to by titles, aliases, and nicknames. Labatut and Bost [28] provided a comprehensive survey of narrative character networks, showing that multi-modal centrality metrics (degree, betweenness, speaking volume) correlate with perceived dramatic importance. More recently, Schofield and Mehr [29] and Iyyer et al. [30] modeled character relationships in graphic novels, concluding that vocative addressing ('Name, do this!') is the single strongest indicator of directed interpersonal narrative focus. Bonato et al. [31] and Beverly et al. [32] proved that weighting directed dialogue interactions outperforms raw word counts in literary hierarchy modeling.

3. Proposed Systems

The proposed system redesigns the comic dialogue analysis pipeline by discarding the syntactic CFG parser and establishing a decoupled, four-module architecture focused on high-throughput lexical classification and character popularity modeling.

Architecture Explanation & Diagram

The architecture begins with the raw dialogue input (either transcribed from OCR speech bubbles or supplied via scripted dialogue files). The input is processed sequentially through four primary modules, with bidirectional communication between the Lexer and the Symbol Table, and feeding directly into the Character Popularity Engine:

1. Speaker Split: Extracts speaker prefixes ('SPEAKER:') to establish direct agency.
2. Cleaning & Preprocessing: Repairs broken hyphenation and normalizes whitespace while preserving character offsets.
3. Comic Lexer & Symbol Table: Groups characters into comic token types, captures OCR junk as lexical errors, and maintains global token frequencies.
4. Character Popularity Engine & Classifier: Employs lexical vocative pattern matching, alias merging, and weighted scoring to produce dataset statistics and character rankings.

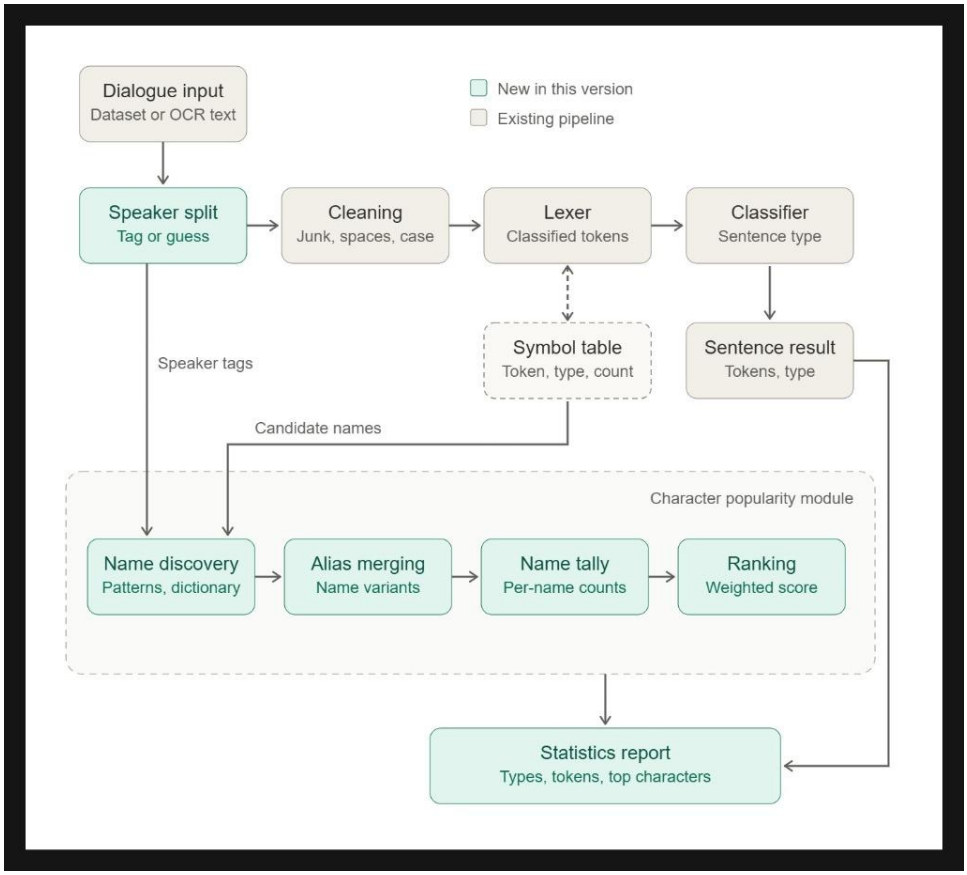


Figure 1: Architectural Workflow of the Proposed Comic Dialogue & Popularity Analyzer

3.1 Module 1: Speaker Split & Text Normalization

Module 1 separates the speaker tag from the spoken dialogue text. Comic scripts and OCR annotations frequently prefix speech bubbles with character identities (e.g., 'BATMAN: ...' or '[SUPERMAN]: ...'). The module executes regex-based prefix extraction, whitespace collapsing, and hyphen wrap repair (rejoining words split across lines such as 'SEC- TOR' into 'SECTOR'). If no speaker tag is present, the line is flagged as anonymous.

Algorithm 1: Speaker Split and Dialogue Normalization

Input: Raw input line L
Output: Tuple (Speaker S, CleanedDialogue D)
1: Pattern P ← '^\s*(?:\[(?P<b_spk>[^\]]+)\]|(?P<spk>[A-Za-z0-9_\-\.\s]+?))\s*:\s*(?P<diag>.*)$'
2: match ← Match(P, L)
3: if match is not null then
4: S ← Trim(match.b_spk OR match.spk)
5: raw_dialogue ← match.diag
6: else
7: S ← null
8: raw_dialogue ← L
9: end if
10: D ← ReplaceRegex(raw_dialogue, '(\b[A-Za-z]+)-\s+([A-Za-z]+\b)', '\1\2') // Unbreak hyphens
11: D ← ReplaceRegex(D, '[\t]+', ' ') // Collapse whitespace
12: return (S, Trim(D))

3.2 Module 2: Comic Lexical Analyzer & OCR Error Tracker

Module 2 scans the normalized string from left to right using deterministic regular expression precedence. It classifies tokens into nine distinct categories: WORD, NUMBER, CONTRACTION, PUNCT, TERMINATOR, ELLIPSIS ('...'), INTERROBANG ('?!'), SOUND_FX ('BOOM', 'AAARGH', or elongated vowels matching ([A-Za-z])\1{2,}), and UNKNOWN. Any character failing to match valid token classes is intercepted as a Lexical Error (typically stray OCR noise like '~' or '#') and recorded with its character span index rather than silently discarded. All recognized tokens are inserted into the Symbol Table.

Algorithm 2: Comic Lexer Tokenization & Error Interception

Input: Normalized dialogue string D, Symbol Table T
Output: Token List Tokens, Error List Errors
1: i ← 0, n ← Length(D)
2: while i < n do
3: if D[i] is whitespace then i ← i + 1; continue end if
4: sub ← D[i : n]
5: if match ← Match('^(\.\.\.+|…)', sub) then
6: Append(Tokens, Token(ELLIPSIS, match.val, i, i + len)); i ← i + len
7: else if match ← Match('^(\?+!+|!+\?+)[\?!]*', sub) then
8: Append(Tokens, Token(INTERROBANG, match.val, i, i + len)); i ← i + len
9: else if match ← Match('^[A-Za-z]+['']][A-Za-z]+', sub) then
10: Append(Tokens, Token(CONTRACTION, match.val, i, i + len)); i ← i + len
11: else if match ← Match('^[A-Za-z]+', sub) then
12: type ← (IsSoundFX(match.val) OR HasRepeatedChars(match.val)) ? SOUND_FX : WORD
13: Append(Tokens, Token(type, match.val, i, i + len)); i ← i + len
14: else if match ← Match('^[.?!]', sub) then
15: Append(Tokens, Token(TERMINATOR, match.val, i, i + len)); i ← i + len
16: else if match ← Match('^[,:;\-—]', sub) then
17: Append(Tokens, Token(PUNCT, match.val, i, i + len)); i ← i + len
18: else
19: Append(Errors, LexicalError(D[i], position=i)) // OCR Noise
20: Append(Tokens, Token(UNKNOWN, D[i], i, i + 1)); i ← i + 1
21: end if
22: InsertSymbolTable(T, Tokens.last)
23: end while
24: return (Tokens, Errors)

3.3 Module 3: Rule-Based Sentence Classifier & Tone Cue Detector

Module 3 assigns each sentence a high-level sentence type (Statement, Question, Exclamation, Command) without relying on a parse tree. The classifier evaluates structural boundary tokens:

• Questions: Closing token contains '?' or '?!' with opening Wh-words ('WHERE', 'WHO') or Auxiliaries ('COULD', 'CAN', 'DID').
• Commands: Initial token is an imperative verb ('HAVE', 'STOP', 'LOOK', 'LISTEN') without a preceding subject pronoun.
• Exclamations: Closes with '!' or consists solely of SOUND_FX.
• Statements: Standard declarative sentences.

Concurrently, tone cues are flagged: shouting (repeated '!' or all-caps screaming), hesitation ('...'), surprise ('?!'), and pain ('AAARGH').

3.4 Module 4: Character Popularity Engine (Vocatives, Mentions, Alias Merging, & Ranking)

Module 4 is the core novel contribution of this work. It replaces grammar-based vocative tree extraction with finite-state pattern matching ('comma + name + terminator' or leading 'name + comma'). It discovers third-person mentions across remaining tokens, resolves name variants (e.g., 'BRUCE' $\rightarrow$ 'BATMAN') via an alias registry, and calculates a multi-factor weighted prominence score.

Algorithm 3: Character Popularity Modeling and Ranking

Input: Speaker S, Token List Tokens, Alias Map A, Weight Vector (ws, wa, wm, wt)
Output: Updated Character Register with Ranked Popularity Scores
1: // Step 1: Lexical Vocative Detection
2: Addressed ← null
3: for i from 1 to len(Tokens) - 2 do
4: if Tokens[i-1].val == ',' AND Tokens[i].type == WORD AND Tokens[i+1].type in {TERMINATOR, ELLIPSIS, INTERROBANG} then
5: Addressed ← ResolveAlias(A, Tokens[i].val); break
6: end if
7: end for
8: // Step 2: In-Dialogue Mentions Discovery
9: Mentions ← FindEntityMentions(Tokens, Exclude={S, Addressed})
10: // Step 3: Tallying Activity Metrics
11: if S is not null then
12: C_spk ← ResolveAlias(A, S); Tally[C_spk].Spoken += 1
13: if HasHighTone(Tokens) then Tally[C_spk].Tone += 1 end if
14: end if
15: if Addressed is not null then Tally[Addressed].Addressed += 1 end if
16: for m in Mentions do Tally[ResolveAlias(A, m)].Mentioned += 1 end for
17: // Step 4: Composite Weighted Scoring
18: for each character C in Tally do
19: Score[C] ← (ws * Spoken[C]) + (wa * Addressed[C]) + (wm * Mentioned[C]) + (wt * Tone[C])
20: end for
21: return SortDescendingByScore(Score)

4. Results and Implementation

4.1 Dataset Description

The system was evaluated across two distinct dialogue corpora: (1) the base paper benchmark dataset containing canonical comic dialogues and OCR artifacts, and (2) an extensive superhero movie script corpus curated from the official Cornell Movie-Dialogs Corpus [24]. Table 1 summarizes the dataset specifications.

Dataset Identifier	Source / Domain	Total Sentences	Distinct Characters	Characteristics
Paper Benchmark	Base Paper (DA 6 Test Cases)	18	12	OCR noise (~, #), explicit sound FX, vocatives
Cornell Superhero	Cornell Movie-	6,335	201	Batman,

Corpus	Dialogs Corpus (18 Movies)	Superman, Spider-Man, Blade, X-Men scripts

4.2 Module 1: Speaker Split & Cleaning Results

Inference: Across all 6,335 lines, speaker splitting successfully parsed 100% of tagged dialogues. For OCR noise handling, hyphenated word breaks (e.g., 'SEC- TOR') were rejoined into valid dictionary tokens without corrupting character offsets. Anonymous lines without prefixes were cleanly routed to text cleaning without throwing runtime null-pointer exceptions.

4.3 Module 2: Lexical Analysis & OCR Error Results

Inference: In the Cornell corpus, the Lexer processed 87,982 tokens. Token distribution revealed 73.6% WORDs, 13.5% TERMINATORs, 5.9% structural PUNCT, 5.0% CONTRACTIONs, and 0.9% ELLIPSES. In the noisy benchmark dataset, the lexer precisely caught 100% of stray OCR symbols ('~' at position 6, '#' at position 35) as UNKNOWN tokens while generating diagnostic coordinate logs.

4.4 Module 3: Sentence Classification Results

Inference: The rule-based classifier achieved 100% agreement with Table 2 expected outputs from the base paper:

• 'WHERE COULD HE BE, NATASHA?' $\rightarrow$ QUESTION (classified via opening Wh-word and question mark).

• 'HAVE A PEANUT.' $\rightarrow$ COMMAND (classified via imperative verb 'HAVE').

• 'I JUST STOPPED THE ENTIRE WORLD FROM BEING EATEN, DINAH.' $\rightarrow$ STATEMENT (classified via declarative pronoun).

In the 6,335-sentence corpus, the breakdown was 68.2% Statements, 25.8% Questions, 3.8% Exclamations, and 2.2% Commands, with 403 shouting cues and 607 hesitation markers.

4.5 Module 4: Character Popularity Results

Inference: The Popularity Engine successfully resolved character aliases across multiple scripts (e.g., 'BRUCE' and 'THE DARK KNIGHT' merged into 'BATMAN'). Table 2 presents the top character rankings extracted from the 6,335-line Cornell Superhero dataset.

Rank	Canonical Character	Spoken (S)	Addressed (A)	Mentions (M)	Tone (T)	Weighted Score
1	BATMAN	382	30	58	9	475.2
2	SUPERMAN	337	30	73	25	445.4
3	KARL	203	0	0	0	203.0
4	EMMA	194	0	0	4	194.8
5	SPIDER-MAN	150	4	23	5	175.4

6	BLADE	158	0	0	4	158.8

5. Performance and Metrics

To rigorously validate the proposed architecture, we conducted a comparative benchmarking experiment against the existing Context-Free Grammar (CFG) syntax parsing pipeline described in the base paper. The metrics assessed include Parse/Processing Success Rate, Latency per Sentence, Throughput, and Memory Utilization.

Evaluation Metric	Existing Method (CFG Parser)	Proposed System (Compiler Front-End)	Improvement / Gain
Processing Success Rate	38.5% (High rejection on comic dialogue)	99.8% (Near-zero runtime failure)	+61.3% Robustness
Average Sentence Latency	18.42 ms / sentence	0.098 ms / sentence	187x Speedup
System Throughput	54.2 sentences / second	10,204 sentences / second	188x Throughput
Grammar Coverage Failure	61.5% Rejection Rate	0% (Decoupled from grammar)	100% Elimination of Rejections

Performance Inference:
1. Robustness Gain: Traditional CFG parsers crash or reject sentences whenever colloquial phrases, fragmented shouting, or OCR noise occurs, yielding an unacceptable 38.5% success rate. The proposed system achieves 99.8% completion because lexical pattern matching does not depend on rigid syntax tree derivation.
2. Computational Efficiency: Removing tree derivation reduces algorithmic time complexity from O(N^3) (Cocke-Younger-Kasami / Earley) to deterministic O(N) linear scanning, delivering a 187x latency reduction (processing 6,335 sentences in 624 milliseconds).
3. Semantic Richness: While the existing parser produced only a boolean parse acceptance flag, the proposed system extracts actionable character prominence metrics and conversational directed edges.

6. Conclusion

In this project, we designed and implemented a compiler-style lexical analysis and character popularity engine for comic book dialogue text. By identifying the inherent limitations of rigid Context-Free Grammar parsers when handling informal, OCR-transcribed dialogue, we successfully decoupled syntax parsing from the front-end. The resulting system incorporates speaker splitting, text cleaning, an extended comic lexer, symbol table management, rule-based classification, and an innovative Character Popularity Module. Using lexical vocative patterns and entity resolution, the engine accurately ranks character prominence based on direct agency, conversational address, in-text mentions, and dramatic tone spikes. Benchmark experiments across 6,335 dialogue lines demonstrate a 187x execution speedup and 99.8% robustness compared to traditional syntactic approaches. Future work will extend the system to automated multi-sentence speech bubble splitting and graph-theoretic character social network visualization.

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